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Parameter matching and multi-objective optimization of final ratio of range-extended electric vehicle

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Abstract: **Range-Extended Electric Vehicle (REEV) is an important technical route to alleviate the range anxiety, and the final ratio of its transmission system is a key parameter affecting the vehicle dynamics and economy. Then, the matrix calculation module is used to scan the final ratio parametrically in the range of 2.1 to 8.9, and the multi-objective optimization is carried out with the weighted comprehensive energy consumption as the optimization objective. The results show that the final ratio has a significant impact on energy consumption, and there is an optimal balance point. Finally, the optimal final ratio is determined to be 4.9. After optimization, the fuel consumption of the vehicle is reduced from 3.64L to 3.46L, and the economy is improved by about 4.9%. Meanwhile, the dynamic performance index is still better than the design requirements. This study provides an effective theoretical method and engineering reference for the matching optimization of REEV transmission system.**

Keywords- REEV; Final ratio; AVL Cruise Simulation; Energy Consumption Analysis

I. INTRODUCTION

REEV is an innovative vehicle that combines a traditional fuel engine with an electric drive system [1]. Its core design concept is to charge the battery pack through a small displacement engine to drive the generator, thereby extending the pure electric range of the vehicle [2].

Compared with pure electric vehicles (BEV), REEV can realize long-distance driving without relying on external charging facilities, effectively solving the problem of "mileage anxiety"; compared with plug-in hybrid electric vehicles (PHEV), REEV engine is not directly involved in driving, only used as "range extender", simplifying the powertrain structure and improving energy utilization efficiency [3]. In the global new energy vehicle market, REEV has become an important solution in the transition stage due to its technical characteristics of "oil and electricity complementarity" [4].

The final ratio is a key parameter in the transmission system, critically affecting vehicle dynamics, economy, and NVH [5]. A larger ratio enhances torque for better acceleration, while a smaller one improves high-velocity efficiency [6]. However, an improper balance can sacrifice power reserve or increase component load, impacting reliability. Despite the REEV's advantages, its performance is limited by a lack of systematic research on final ratio optimization across full lifecycle and multiple scenarios, beyond single condition or index studies [7, 8].

This study analyzes the impact of the final drive ratio on the performance and fuel economy of a REEV through simulation. It aims to establish quantitative relationships, propose a multi-objective optimization design method to balance these competing goals, and validate the optimized solution in real driving scenarios. The findings offer theoretical support for REEV transmission design, promoting more efficient and low-carbon vehicle technology.

II. REEV STRUCTURE AND SIMULATION MODEL

The studied REEV employs a series hybrid architecture, comprising a 150kW drive motor, a 1.5L range extender, and a 40.5kW•h battery pack [9]. The range extender activates only to charge the battery when the State of Charge (SOC) is low [10]. This "primarily electric drive, supplemented by power generation" approach effectively alleviates range anxiety and features simpler control logic than Plug-in Hybrid Electric Vehicles (PHEVs).

In the simulation modeling process, this study set the following idealized conditions to simplify computational complexity and focus on power system matching analysis: excluding the impact of accessory loads such as air conditioning and power steering on energy consumption; Set the ambient temperature constant at 25 ° C, ignoring the impact of extreme temperatures on battery and engine performance; Adopting standard driving cycle tracking strategy without considering actual driving behavior differences; Assuming that the road slope and wind resistance coefficient are constant values, the actual dynamic changes in road conditions were not simulated. These assumptions may lead to simulation results that outperform actual road test performance, but they provide a clear benchmark analysis framework for system parameter optimization. The overall model is shown in Figure 1.

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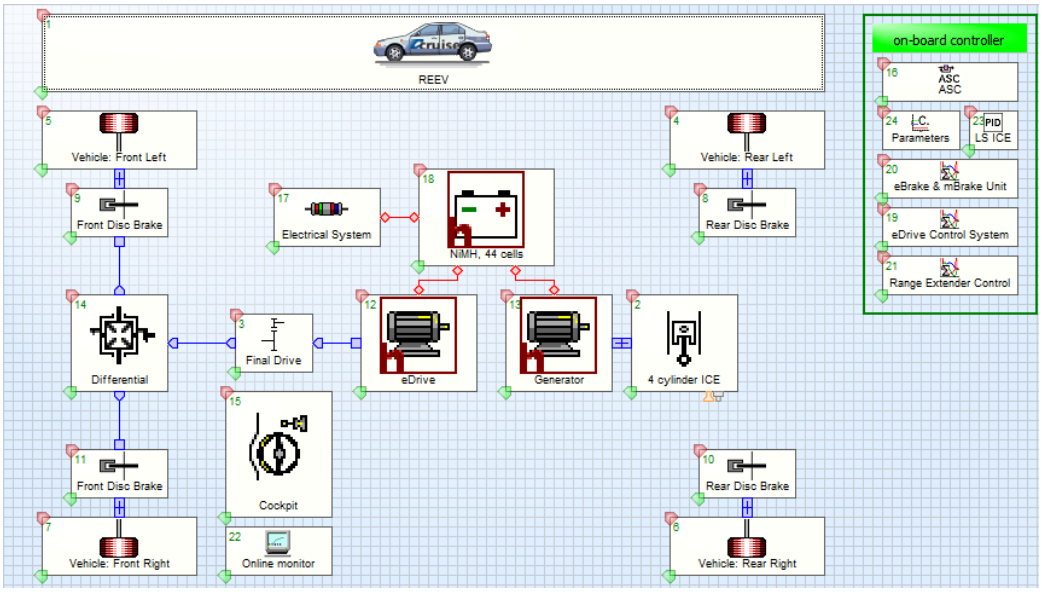


Figure 1. AVL Cruise REEV Simulation

III. Simulation Results and Analysis

A. Simulation results

Simulation curves can directly reflect the dynamic response characteristics and energy flow process of vehicles, which is an important basis for performance evaluation.

As shown in Figure 2, the simulated vehicle velocity closely follows the target driving cycle, demonstrating rapid torque response and stable tracking across both low and high-velocity phases. This accurate velocity matching validates the powertrain model's parameters and control strategy, ensuring reliable results for subsequent economic simulations.

The battery SOC management is central to the REEV's energy strategy. As shown in Figure 3, SOC decreases from 90% under battery power until it reaches a 30% threshold, triggering the range extender. The range extender then stabilizes the SOC, verifying the control strategy's effectiveness in preventing battery depletion and achieving the extended-range function.

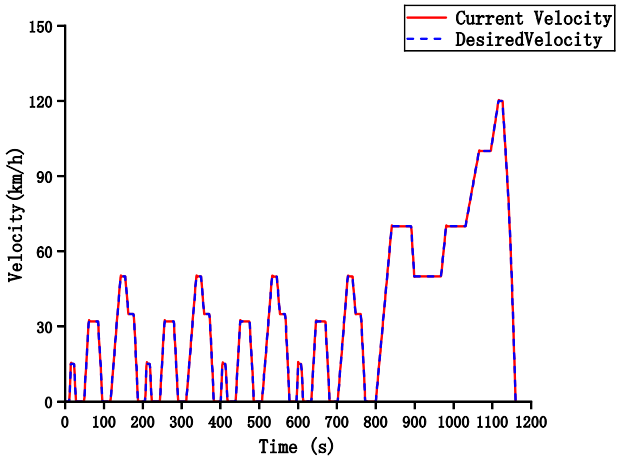


Figure 2. Velocity following curve

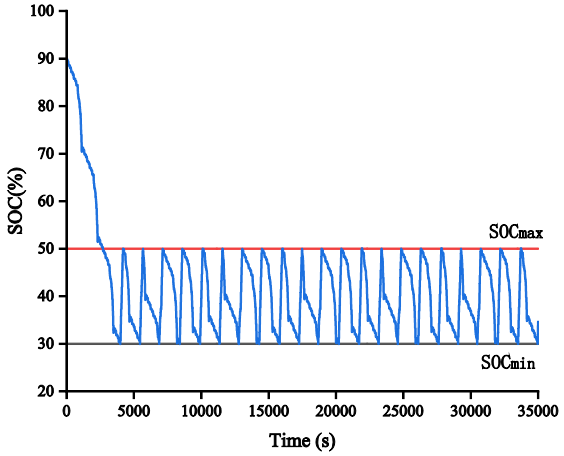


Figure 3. Change curve of SOC with time

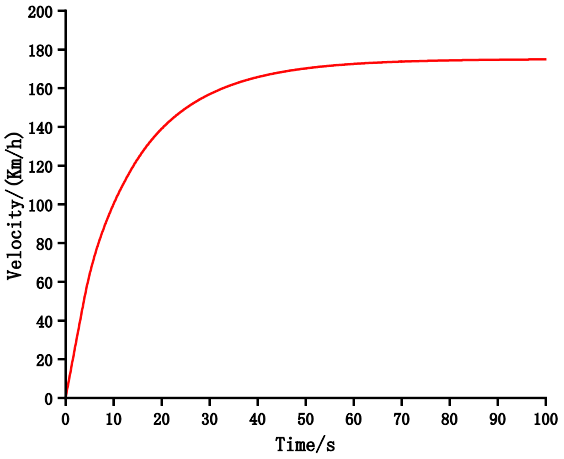


Figure 4. Time varying curve of vehicle velocity

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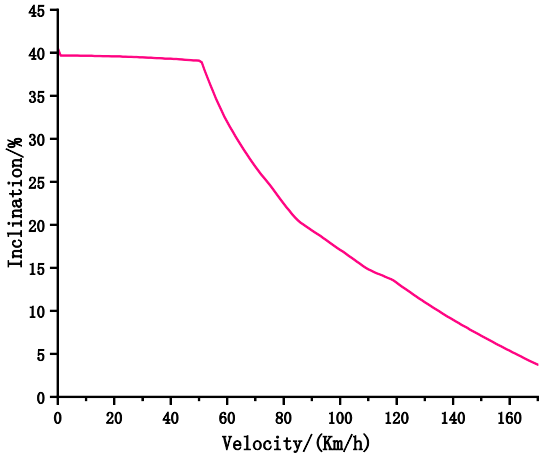


Figure 5. Curve of MAX gradient varying with velocity

Figure 4 shows the vehicle's acceleration and top velocity performance. The 0-100 km/h time of 9.97 seconds and a top velocity of 174 km/h both significantly exceed the design targets (15s and 140 km/h), demonstrating strong motor torque output and sufficient power reserve at high velocity.

Figure 5 evaluates the vehicle's climbing ability, showing a maximum grade of 40.57% at low velocity, far exceeding the 25% target due to high motor torque. Climbing capacity decreases with velocity as more power is used to overcome resistance. This result also suggests potential to adjust the final ratio to optimize high-velocity energy consumption while meeting basic climbing requirements.

B. Summary and analysis of simulation results

The simulation results of key performance indicators were summarized and compared with the design requirements, as shown in Table 1. Comparative analysis showed that the simulation results were significantly better than the design objectives in all four core indicators.

TABLE 1. SIMULATION RESULT

Parameters	Unit	Simulation result	Requirement
Fuel consumption per hundred kilometers	L	3.64	7
Maximum gradeability	%	40.57	25
Acceleration time per hundred kilometers	S	9.97	15
maximum velocity	km/h	174	140

A significant gap exists between simulation results and design targets, revealing "excess performance" across all metrics. Fuel consumption (3.64 L/100km) is substantially lower than the design value (7 L/100km), attributed to idealized simulation conditions where the range extender operates optimally and accessory loads are neglected. Meanwhile, dynamics indicators—maximum gradeability (40.57%), 0–100 km/h acceleration (9.97 s), and top velocity (174 km/h)—all significantly exceed requirements, indicating ample power reserve and a final ratio biased toward dynamic performance.

IV. FINAL RATIO OPTIMIZATION ANALYSIS

A. Parametric optimization design based on matrix calculation

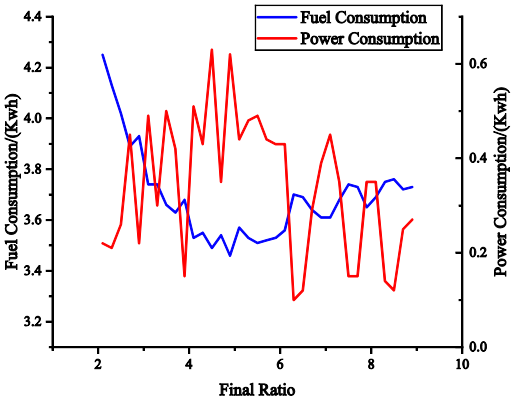


Figure 6. Curve of Fuel Consumption and Power Consumption with Final ratio

This study employs AVL Cruise's matrix calculation module to parametrically scan the final drive ratio from 2.1 to 8.9 (increment: 0.2), evaluating 35 schemes under a unified driving cycle. As shown in Figure 6, energy consumption exhibits a nonlinear relationship with the final ratio. Excessively small ratios increase fuel consumption due to inefficient motor and range extender operation under high-torque conditions, while overly large ratios raise energy use by reducing motor efficiency and increasing mechanical losses during high-velocity cruising. An optimal ratio range balances torque output and component efficiency, minimizing overall energy consumption.

B. Decision making based on weighted comprehensive energy consumption

In order to quantitatively evaluate the overall energy efficiency under different final ratio schemes and make final decisions, this study defines weighted comprehensive energy consumption (comprehensive energy consumption) as the core evaluation index, which is consist of fuel consumption *0.9+ power consumption *0.1. Considering that the REEV model still uses fuel as the main energy source in actual use, its calculation formula is set as follow. This formula focuses more on improving fuel economy. Calculate the combined energy consumption for all final ratio and plot it as shown in Figure 6.

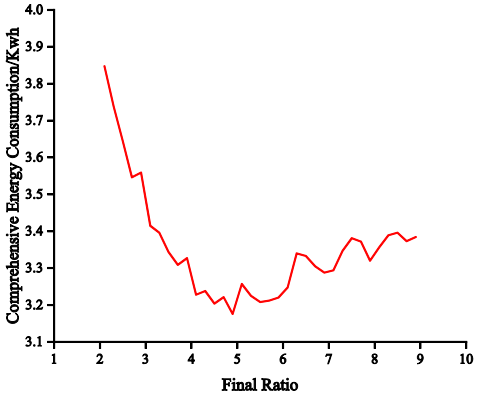


Figure 7. The curve of comprehensive energy consumption changing with the final ratio

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Figure 7 shows that the weighted comprehensive energy consumption curve presents a very obvious "U" shape, and there is a global minimum point. The corresponding final ratio value is the theoretical optimal solution after weighing the global energy consumption. Through optimization, the optimal final ratio for minimizing the comprehensive energy consumption is determined to be 4.9.

C. Validation and comparison of optimization results

The optimal final ratio of 4.9 is substituted into the model for verification simulation, and the key performance indicators are compared with those before optimization and design requirements. The results are summarized in Table 2.

TABLE 2 RESULTS AFTER OPTIMIZATION

Parameters	Unit	Before simulation	After simulation
Fuel consumption per hundred kilometers	L	3.64	3.46
Maximum gradeability	%	40.57	28.58
Acceleration time per hundred kilometers	S	9.97	11.68
Maximum velocity	km/h	174	185

The optimization successfully reduced 100-km fuel consumption from 3.64L to 3.46L (4.9% improvement) while maintaining dynamic performance above design targets - maximum gradeability at 28.58% (target: 25%) and 0-100 km/h acceleration at 11.68s (target: 15s). The increased top velocity further confirms preserved high-speed capability. This matrix-based approach effectively converts surplus dynamic performance into economic gains, providing a scientific foundation for REEV powertrain design.

D. Potential impact analysis on NVH performance

The final ratio significantly influences vehicle NVH performance. A larger ratio may increase high-frequency noise due to elevated motor speeds, while a smaller one could induce engine vibration under inefficient high-load conditions. The optimal ratio of 4.9 determined in this study ensures power performance while avoiding resonance zones of the motor and engine, thereby improving vehicle smoothness. Future work should incorporate NVH simulation or testing to quantitatively evaluate the impact of this ratio on noise and vibration.

V. CONCLUSION

This study optimizes the final drive ratio of a Range-Extended Electric Vehicle (REEV) through AVL Cruise-based simulation. Parametric sweep analysis (2.1–8.9 ratio range) identified an optimal ratio of 4.9, reducing fuel consumption by 4.9% (3.64→3.46 L/100 km) while improving gradability. The results demonstrate the effectiveness of simulation-driven optimization for REEV transmission design, providing a practical reference for efficient vehicle development.

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